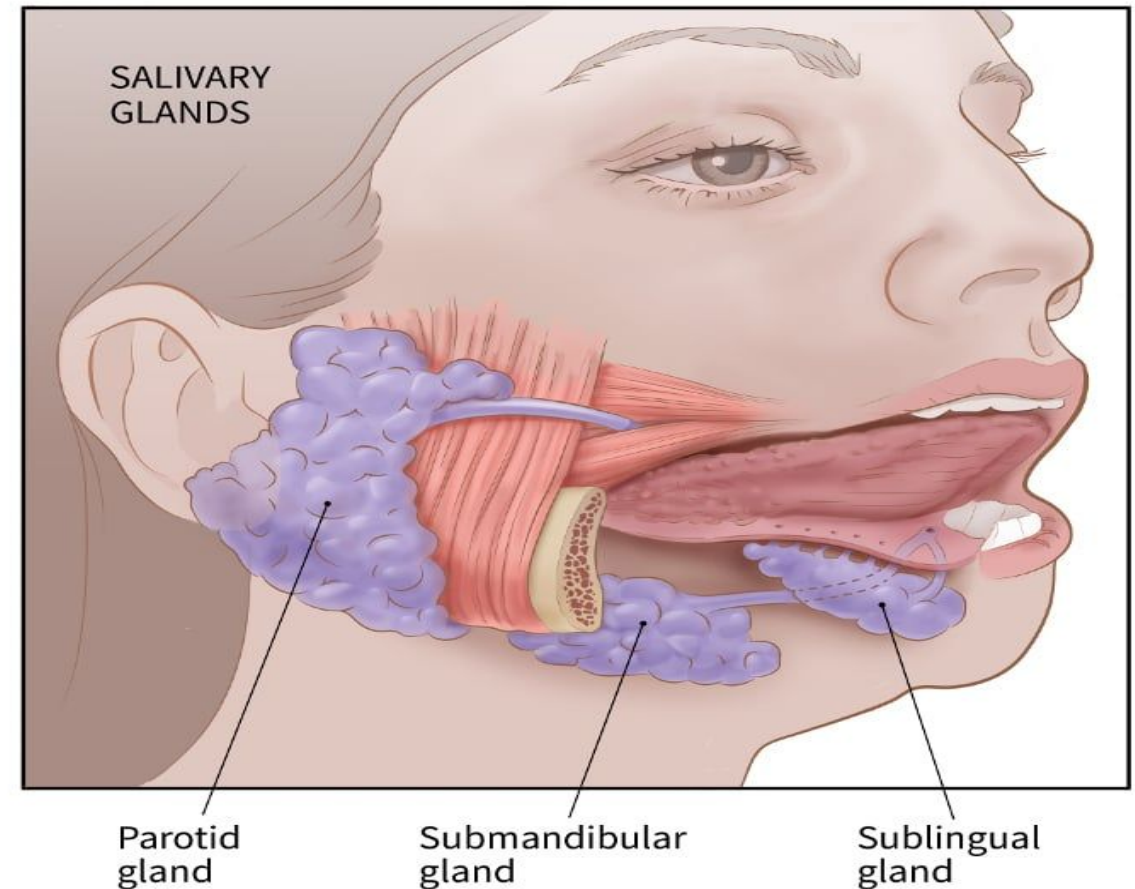


To test the salivary functions of a subject.



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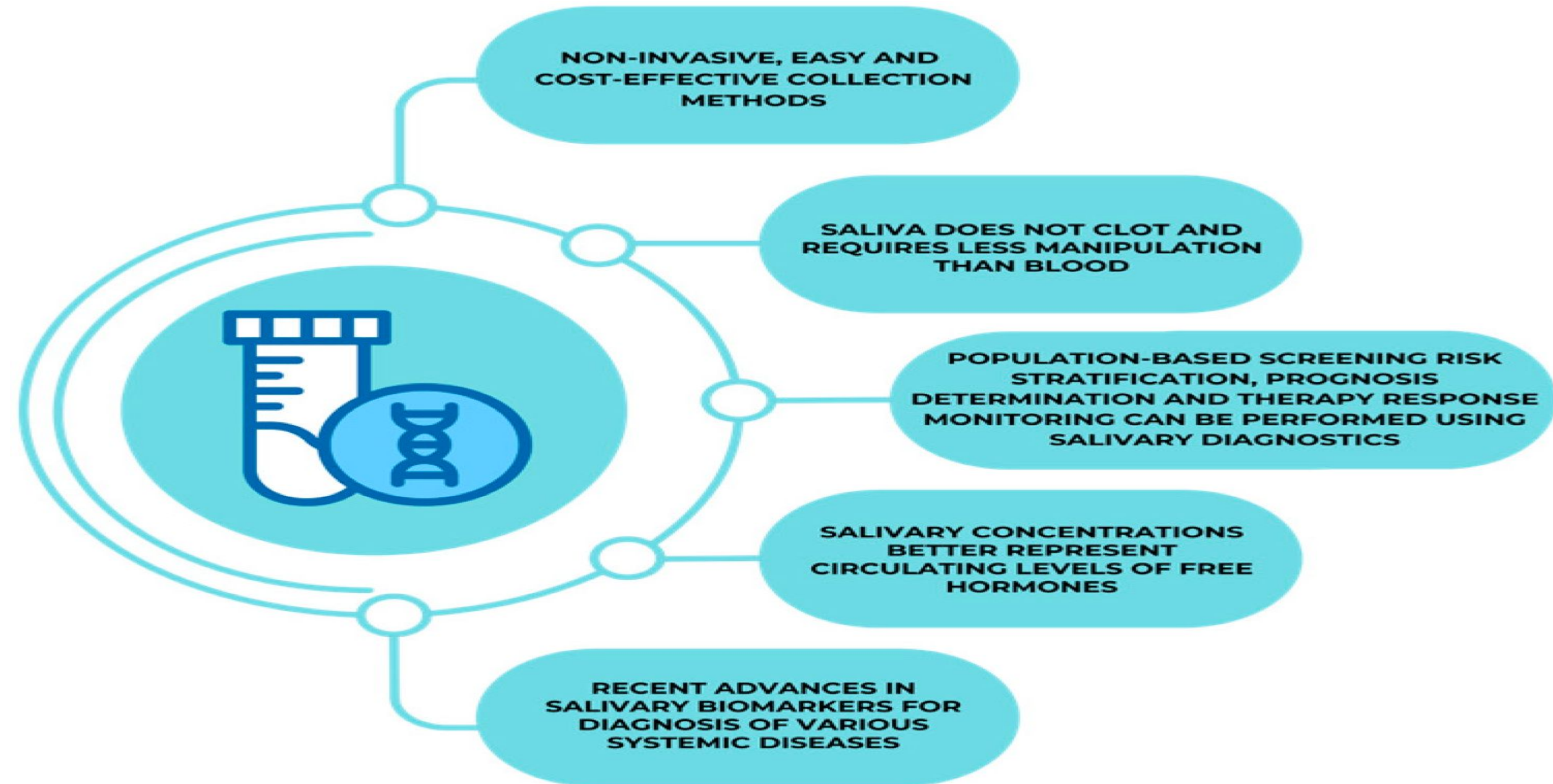
REQUIREMENTS:

- Stop watch,
- pH paper,
- Vitamin C tablets,
- 10 cc disposable syringe,
- 2 cc disposable cups,
- tissue paper,
- and unflavored paraffin.

Functions of Saliva



Saliva as a Diagnostic Tool for systemic Disease



THEORY:

Saliva is vital to the integrity of the teeth as well as the soft tissues. It is not uncommon to observe patients presenting with loss of tooth structures due to dental caries or dental erosion.

The most possible reason for change in the oral balance that is favoring demineralization may be answered by measuring important salivary parameters.

1. Hydration status testing:

Measures salivary production.

Everting the lower lip and observing the droplets of saliva formed in one minute will assess the hydration status of unstimulated saliva. (> or < 30 seconds)

2. Salivary Consistency testing: assesses the viscosity of unstimulated saliva.

Frothy or bubbly saliva suggests increased viscosity. This may be due to reduced water content in the saliva.

3-Saliva pH testing: measures the pH of the unstimulated saliva.

The normal salivary pH is 7.0 – 7.5 or neutral to slightly alkaline. With salivary gland hypo function, the salivary pH decreases. Erosion of dental hard tissue occurs at pH 5.5. A low salivary pH contributes to mucosal discomfort.

4-Stimulated Saliva pH testing: The alkaline levels of saliva should rise after chewing vitamin C or eating a meal, indicating a good mineral reserve against acidic diet and other environmental factors.

5-Normal salivary flow rate (hydration status and stimulated saliva flow rate) imparts a strong protective effect against dental caries. Significantly lower salivary flow rate may be associated with a number of predisposing factors such as lack of raw material (water), lack of stimulus to the salivary gland or could be a problem with salivary gland itself.

6.Stimulated Saliva Flow testing: measures the quantity of saliva produced in 5 minutes while chewing an unflavored piece of wax.

7. Salivary buffering capacity protects the tooth from dental caries, neutralizing acid in oral cavity after sugar intake, thereby preventing reduction in salivary pH.

This also depends upon extrinsic factors such as dietary and oral hygiene habit, as well as intrinsic factor such as bicarbonate content.

Since saliva provides a general protective effect, clinically significant decrease in salivary functions can be considered as an etiologic factor that contributes to the development of dental caries.

If only conventional dental treatment is done in these subjects without treating the defect in the salivary parameters, the dentist is sure to see recurrence of dental caries in a relatively short time.

TEST 1: Hydration status testing:

1. With one hand, gently pull down your lower lip in front of a mirror. You will need a good light.
2. With the other hand, dry the inside of your lip with a clean tissue paper.
3. The time taken to form droplets of saliva on the minor glands of the lower lip can then be timed.
4. Within one minute, you should notice small beads of saliva on the previously dried surface, leaving a glistening surface.
5. If after one minute it is still dry, you may have a salivary flow problem.

TEST 2: To assess viscosity of saliva:

1. Expectorate all the resting saliva from your mouth into a small 2 cc disposable cup.
2. The viscosity of saliva is determined by visual examination.
3. Is it watery, bubbly, or frothy and sticky?
4. This is more easily seen by placing the cup against a dark background.

TEST 3: To determine pH of unstimulated saliva:

1. Tear off a small one to two inch piece of the pH paper from the dispenser.
2. Dip the pH test paper directly into the sample of oral fluid collected in test 2, and then remove immediately (should only take one to two seconds).
3. Notice that it is changing color. Compare the color to the chart located on the side of the dispenser. Find the color on the chart that most closely matches the color of the pH paper. Disregard any color changes that continue after 30 seconds.
4. Note the pH number on the chart that corresponds with the color of your pH strip. That is your saliva's pH.

TEST 4: To determine unstimulated salivary flow rate:

1. Fold the needle of 10 cc syringe on itself, recap it, and take out the plunger.
2. Sit on the chair, feet flat on the floor, leaning forward, resting your elbows and arms on your thighs and knees while bending the neck.
3. Hold the open end of syringe, gently to the face.
4. Open your lips and allow saliva to flow passively into the syringe for 5 minutes.
5. Measure the total volume collected in 5 minutes (do not include the foam), & divided by five, to get the salivary flow rate / minute – “unstimulated”.

5: To determine stimulated salivary flow rate:

1. Chew a piece of unflavored paraffin (1" by 1") for 1 minute.
2. Swallow all the saliva in the mouth then immediately start the timer. (Set for 2 minutes).
3. Repeat the measurement process as in test 4, while continuing to chew for the entire 2 minute collection period, with head tilted down.
4. Measure the volume of saliva in the syringe and divide by 2 to get the stimulated salivary flow rate.
5. Repeat the measurement process as you did in test 1 above.

Note: A 1% topical solution of citric acid applied to the tongue can also be used to stimulate the saliva.

TEST 6: To determine pH of stimulated saliva:

1. Place a chewable Vitamin C tablet in your mouth and suck on it until the entire mouth is coated, then spit out the tablet.
2. Swallow four times.
3. Check the pH of your saliva as you did in test 1 above.

Observations:

	Normal	Abnormal	Observation
Hydration Status	< 30 sec	> 30 sec	
Viscosity	Watery, clear	Frothy, stringy	
Salivary pH – unstimulated	7.0 – 7.5	< 6.5	
Salivary pH – stimulated	Increases (write pH observed)		
Salivary flow rates – unstimulated	0.3 ml/min	< 0.1 ml/min	
Salivary flow rates – stimulated	1 – 2 ml/min	< 0.5 ml/min	

PRECAUTIONS:

1. The pH of saliva is affected by digested activity. Don't smoke, eat, drink, chew gum, or brush your teeth for at least two hours.
2. Hydration should be tested in good light.
3. Refrain from swallowing when testing for the salivary flow rate.
4. While measuring salivary flow in the syringe, do not measure the foam.

Relevant Question

Q-1-What is the composition and function of Saliva

Q-2-What is the pH of saliva?

Q-3- Why is it important to test Saliva after meals/vitamin C

Q-4-What are the causes of decreased salivary flow rate and its effect on saliva functions?

Q-5-How can you increase your salivary flow (hydration)?

Q-1-What is the composition and function of Saliva

Saliva is an exocrine secretion primarily composed of 99% water and 1% essential substances like electrolytes, mucus, and enzymes. Produced daily by the major salivary glands, its primary functions are to lubricate the mouth, aid in swallowing, begin digesting carbohydrates, and protect against oral bacteria.

Function

Saliva performs several vital roles to maintain your health:

Digestion: It moistens food so it is easy to swallow and initiates the chemical breakdown of carbohydrates and lipids before food reaches the stomach.

Oral Protection: The mucus coats the delicate lining of your mouth and throat, preventing friction and damage during eating and speaking.

Cavity Prevention: Its electrolytes (specifically bicarbonate) neutralize acids in the mouth, while calcium and phosphate help remineralize tooth enamel to prevent decay.

Immune Defense: Antimicrobial proteins actively lyse (dissolve) bacteria and fight off pathogens, while constant saliva flow physically washes away food debris that fosters bacterial growth.

Taste Enhancement: Saliva dissolves food molecules, allowing them to bind to receptors on the taste buds, facilitating the sense of taste.

Q-2-What is the pH of saliva?

The normal pH of saliva ranges from **6.2 to 7.6**, which is slightly acidic to slightly alkaline. An optimal neutral state is considered to be around **7.0 to 7.2**.

Q-3- Why is it important to test Saliva after meals/vitamin C

Testing your saliva after eating or taking Vitamin C is important because these introduce foreign substances and sudden pH shifts that can skew health diagnostics.

What are the causes of decreased salivary flow rate and its effect on saliva functions?

Causes of Decreased Salivary Flow Rate

Salivary gland hypo function is driven by a variety of medical, lifestyle, and environmental factors:

Medications: The most common cause. Hundreds of prescription and over-the-counter drugs (e.g., antihistamines, antidepressants, antihypertensive, and antianxiety medications) interfere with the nerve-mediated stimuli required for saliva production.

Radiation Therapy: High-dose radiation directed at the head or neck can irreversibly damage the salivary gland acinar cells, significantly hindering their ability to produce saliva.

Autoimmune Diseases: Conditions like Sjögren's disease cause chronic inflammation where the body's immune system attacks moisture-producing glands, profoundly reducing saliva and tear production.

Systemic Health Conditions: Uncontrolled diabetes, HIV/AIDS, Parkinson's disease, Alzheimer's disease, and chronic kidney disease can disrupt neurological reflexes or cause systemic dehydration that impacts glandular output.

Dehydration & Lifestyle: Insufficient fluid intake, mouth breathing, and the use of tobacco, alcohol, or cannabis also significantly diminish saliva production.

Effect on Saliva Functions

Effect on Saliva Functions

A diminished salivary flow rate directly impairs the three primary functions of saliva:

1. Protection & Buffering

- **Enamel Demineralization:** Saliva continuously clears sugars, washes away food debris, and neutralizes bacterial acids. Without it, acids remain on the teeth, leading to a surge in dental caries (cavities) and demineralization.
- **Oral Infections:** Saliva contains antimicrobial agents and immunoglobulins (like sIgA and histatins) that prevent overgrowth of bacteria and fungi. A lack of these agents leads to a higher incidence of oral candidiasis (thrush) and other bacterial infections.

2. Lubrication

- **Tissue Friction:** Saliva provides a protective, lubricating film (via mucins) over the oral mucosa and lips. Reduced flow causes painful oral burning, peeling or cracking of the lips, and a rough tongue.
- **Prosthesis Irritation:** A lack of lubrication makes it very difficult for individuals to retain removable dentures, frequently causing denture sores.

3. Digestion & Sensation

- **Impaired Taste:** Saliva acts as a solvent for taste molecules, carrying them to the taste buds. Decreased flow leads to an altered or diminished sense of taste.
- **Dysphagia (Swallowing) and Speaking:** Reduced moisture makes it difficult to form a food bolus, chew, and swallow, and interferes with the tongue's ability to articulate words smoothly.

5-How can you increase your salivary flow (hydration)?

To increase your salivary flow, consistently sip on water and fluids throughout the day rather than chugging large amounts at once. For maximum benefit, target 2 to 3 liters of daily fluid intake and complement these habits with saliva-stimulating tricks and environmental adjustments.

- Use the following strategies to boost your hydration and salivary flow:
- **1. Mechanical & Sensory Stimulation**
- **Chew Sugar-Free Gum:** The physical act of chewing and the flavor of the gum directly signal your salivary glands to produce more. Opt for gums containing xylitol, which also helps prevent cavities.
- **Suck on Sugar-Free Lozenges:** Citric, mint, or cinnamon-flavored sugar-free candies or lozenges can naturally trigger saliva production.
- **Eat Crunchy Foods:** Snacking on raw, high-fiber, and moisture-rich foods (e.g., apples, carrots, celery, and cucumbers) requires thorough chewing, which stimulates salivary flow.

2. Dietary Adjustments

- **Limit Dehydrating Culprits:** Cut back on caffeine (coffee, tea, soda), alcohol, and tobacco, as these are well-known diuretics that dry out the mouth.
- **Eat Your Water:** Consume water-dense fruits and vegetables like watermelon, strawberries, oranges, and broths to increase your systemic hydration.

3. Environmental & Home Care

- **Use a Humidifier:** Place a cool-mist room humidifier near your bed, especially while sleeping, to add moisture to the air and prevent waking up with a parched mouth.
- **Breathe Through Your Nose:** Mouth-breathing rapidly evaporates moisture and worsens dryness.

4. Over-the-Counter Solutions

- **Use Artificial Saliva Products:** Explore over-the-counter saliva substitutes and dry-mouth mouthwashes (e.g., Biotene or XyliMelts) designed to lubricate the mouth.
- **Use Moisturizing Oral Sprays:** Keep a water or glycerin spray nearby to mist the inside of your mouth when it feels dry.